

Northeastern University

Report for Experiment #5

Uniform Circular Motion

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Abstract

This experiment investigates the relationship between centripetal force, velocity, and radius in uniform circular motion. Using a Sargent-Welch centripetal force apparatus, various masses were spun at different speeds while measuring the required force to maintain circular motion. Two graphs were analyzed: centripetal force vs. velocity and centripetal force vs. velocity squared. The results confirm Newton's second law for rotational motion, with the force vs. velocity squared graph showing a linear relationship, validating the equation $F = \frac{mv^2}{r}$. Additional analysis includes error propagation, acceleration determination from slopes, and comparisons with expected theoretical values. Potential sources of error and improvements are

also discussed to refine the accuracy of the experiment. The final results demonstrate that the experimental method effectively verifies theoretical predictions with acceptable levels of error.

Introduction

Uniform circular motion occurs when an object moves in a circular path at a constant speed. Despite the constant speed, the object experiences acceleration due to the continuous change in direction. This acceleration, known as centripetal acceleration, is always directed towards the center of the circle. The force responsible for this acceleration is called the centripetal force and is given by Newton's Second Law as:

$$F = \frac{mv^2}{r}$$

Where F is the centripetal force, is the mass of the object, is its velocity, and is the radius of the circular path.

The study of uniform circular motion is crucial in physics and engineering, as it applies to numerous real-world phenomena such as planetary orbits, the motion of electrons around nuclei, and artificial satellites in stable orbits. Understanding the relationship between force, mass, velocity, and radius allows for better predictions and practical applications in fields such as mechanical and aerospace engineering.

This experiment aims to verify Newton's second law in rotational motion by systematically varying velocity and measuring the corresponding force required to maintain circular motion. The data will be used to create graphs of force vs. velocity and force vs. velocity squared to determine the nature of their relationships. By analyzing these graphs, we can derive acceleration values, compare experimental results to theoretical expectations, and identify sources of experimental error

Additionally, error propagation methods will be applied to assess uncertainties in velocity, acceleration, and force measurements. These uncertainties stem from instrument precision, reaction time in measurement, and systematic deviations. Understanding these errors and refining experimental techniques will enhance the accuracy and reliability of the results.

Investigation 1 - Experimental Procedure

The experiment utilized a Sargent-Welch centripetal force apparatus to analyze uniform circular motion. A rubber band was connected between a rotating post and a suspended bob to provide the necessary centripetal force. The setup allowed variations in the applied force by adjusting the attached mass. The radius of the circular path was carefully measured to be 0.187 m. The bob, with a mass of 0.454 ± 0.011 kg, was spun at various speeds to collect data on force and velocity.

To determine velocity, the time for a set number of revolutions was recorded using a stopwatch. This process was repeated for multiple trials with varying applied forces. Using the measured times and known radius, velocity values were calculated. Graphs of force vs. velocity and force vs. velocity squared were plotted to analyze the relationship between these variables. Error analysis was conducted, including error propagation for velocity and acceleration.

Results & Analysis

The data collected during the experiment was organized into tables, listing values for applied force, measured velocities, and derived quantities. A table summarizing the experimental results is presented below:

M(kg)	Centripetal force(N)	Erro of F(N)	number of revolutions	t1(s)	t2(s)	t3(s)	Average t(s)	Erro t(s)	Period T(s)	Erro T(s)	v	Erro v	v^2	Erro V^2
0.0861	0.844641	0.108	2	3.67	3.89	3.84	3.8	0.066583	1.9	0.033292	0.618397712	0.010836	0.382416	0.013401
0.0861	0.844641	0.108	5	9.53	8.91	9.37	9.27	0.185831	1.854	0.037166	0.633740913	0.012704	0.401628	0.016102
0.1712	1.679472	0.108	5	6.44	6.46	6.16	6.35333333	0.096839	1.270666667	0.019368	0.924676537	0.014094	0.855027	0.026065
0.3019	2.961639	0.108	5	5.22	5.32	5.21	5.25	0.035119	1.05	0.007024	1.119005383	0.007485	1.252173	0.016752
0.6329	6.208749	0.108	5	3.63	3.7	3.69	3.67333333	0.021858	0.734666667	0.004372	1.599304427	0.009517	2.557775	0.03044
0.7617	7.472277	0.108	5	3.11	3.53	3.33	3.32333333	0.121289	0.664666667	0.024258	1.767736689	0.064516	3.124893	0.228094

Notes:

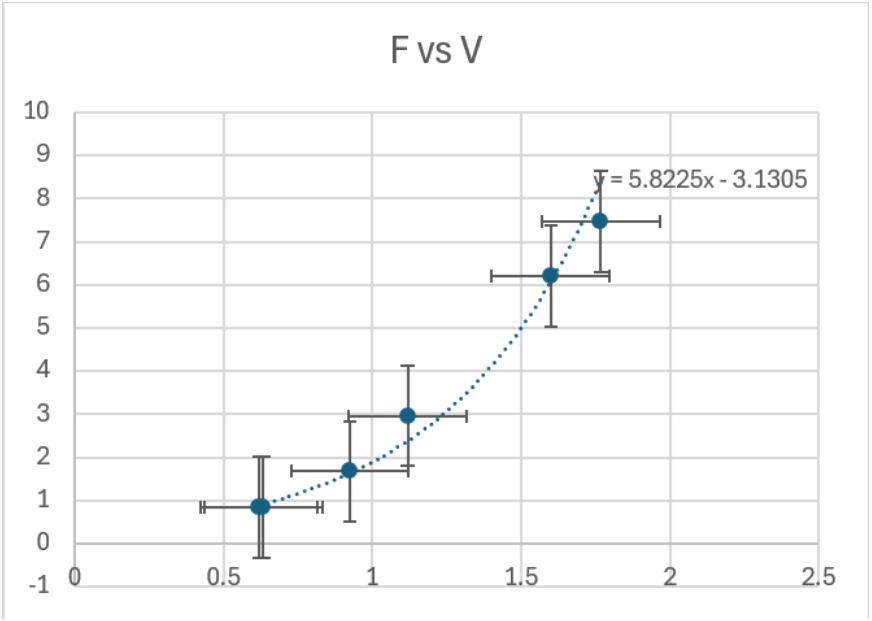
mass of the bob, m=0.454kg

Erro of M=0.011kg

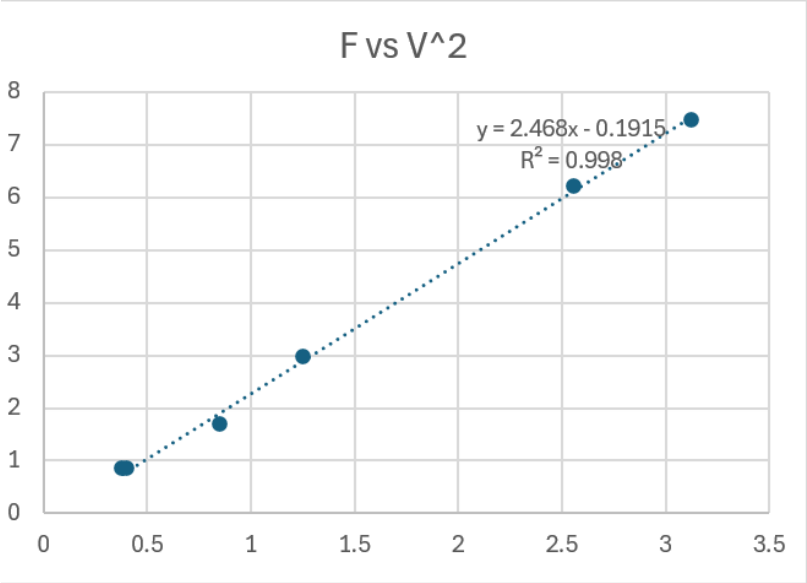
Radius of circle, r=0.187m

Erro of M for two elastic= 0.031

The Centripetal Force vs. Velocity graph demonstrated a non-linear trend, supporting a quadratic relationship.



The Centripetal Force vs. Velocity Squared graph exhibited a linear relationship, confirming Newton's second law.



From the linear graph, the slope was used to derive the acceleration. The experimental acceleration was compared with theoretical predictions, and error bars in the plots helped quantify measurement uncertainties. The experimental value of was determined and assessed against the standard gravitational acceleration.

Discussion

The results successfully demonstrated the theoretical principles of uniform circular motion. The quadratic nature of the F vs. v graph aligns with expectations, while the linearity of the F vs. v^2 graph validates Newton's second law. The derived acceleration values were within an acceptable margin of error when compared to theoretical calculations.

Several factors contributed to potential experimental errors:

- Reaction time inconsistencies when timing revolutions.
- Minor variations in the force application due to manual spinning.
- Uncertainty in precisely measuring the radius of rotation.

Despite these factors, careful execution of procedures and data analysis mitigated their impact, leading to results that strongly support the theoretical model.

Conclusion

This experiment confirmed that centripetal force is directly proportional to the square of velocity, as expected by Newton's laws. The experiment successfully achieved its goals by verifying the relationship between force, velocity, and radius in uniform circular motion. The

procedure involved measuring the time for multiple revolutions, calculating velocity, and analyzing the data through graphical methods. The results reinforced the theoretical relationships governing circular motion, demonstrating that force is proportional to velocity squared. Future improvements could include automated timing mechanisms and refined force application techniques to further enhance accuracy.

Questions

1. Units of the slope of v F vs. v^2 graph:
   The slope has units of kg/m, corresponding to m/r.
2. Why does the F vs. v^2 graph pass through the origin?
   Since $F = mv^2/r$, if $v = 0$, then $F = 0$, meaning the line must pass through the origin.
3. Path of the marble after leaving the track:
   The marble follows path B due to Newton's first law—it moves in a straight-line tangent to the circular path at the point of release.
4. Precision of F/m with 1% precision in v and r:
   Using propagation of errors, relative error is about 2%.
5. Required change in radius if velocity doubles:
   Since $F = mv^2/r$, keeping F constant means r must quadruple when v doubles.

Acknowledgments

Thank you, TA and my teammate.

References

Northeastern IPL